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1**Adrenal medulla**

COMP

You should be able to: Explain why the adrenal medulla is a modified sympathetic ganglion and compare the duration of its circulating catecholamine effect with direct sympathetic innervation.

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2**Pacemaker potential**

COMP

You should be able to: Explain the unstable pacemaker potential of the sinoatrial node by the funny current, calcium entry, and potassium exit and state what sets intrinsic heart rate.

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3**Contractile cell action potential**

COMP

You should be able to: Diagram the ventricular action potential and identify the ion movement responsible for the rapid upstroke, the plateau, and repolarization.

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4**Conduction system**

COMP

You should be able to: Trace an impulse through the sinoatrial node, internodal pathways, atrioventricular node, bundle of His, bundle branches, and Purkinje fibers and explain the purpose of the atrioventricular delay.

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5**Cardiac refractory period**

COMP

You should be able to: Explain why the long refractory period of cardiac muscle prevents tetanus and why that matters for pump function.

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6**ECG waves and intervals**

COMP

You should be able to: Label the P wave, QRS complex, T wave, PR interval, and QT interval and state the electrical event each represents.

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7**ECG interpretation**

COMP

You should be able to: Calculate heart rate from an ECG strip and identify sinus rhythm, tachycardia, bradycardia, and a first degree or complete heart block.

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8**ECG recording**

COMP

You should be able to: Record a lead II ECG, measure the intervals, and correlate each wave with the mechanical event that follows it.

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9**Cardiac cycle**

COMP

You should be able to: Order the phases of the cardiac cycle and state the pressure relationship that opens and closes the atrioventricular and semilunar valves in each phase.

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10**Pressure volume loop**

COMP

You should be able to: Label the four phases of a ventricular pressure volume loop and predict how the loop changes with increased preload, increased afterload, or increased contractility.

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11**Heart sounds**

COMP

You should be able to: Explain the origin of the first and second heart sounds and relate a murmur to valve stenosis or regurgitation by its timing.

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12**Stroke volume and ejection fraction**

COMP

You should be able to: Calculate stroke volume and ejection fraction from end diastolic and end systolic volumes and interpret a reduced ejection fraction.

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13**Cardiac output**

COMP

You should be able to: Calculate cardiac output and cardiac reserve and predict the effect of a change in heart rate or stroke volume on each.

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14**Frank Starling relationship**

COMP

You should be able to: State the Frank Starling law and explain how venous return and end diastolic volume determine the force of the next contraction.

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15**Preload afterload and contractility**

COMP

You should be able to: Distinguish preload, afterload, and contractility and predict how a change in each alters stroke volume.

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16**Autonomic regulation of the heart**

COMP

You should be able to: Explain how sympathetic and parasympathetic input alter heart rate and contractility at the receptor and second messenger level.

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17

Heart sound and pulse correlation

COMP You should be able to: Auscultate the heart sounds, locate the valve areas, and correlate the sounds with the pulse and with the cardiac cycle phases.

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